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Course: Calculus Beta

Assignment: MyLab Math - Uge 5

The first term of a sequence along with a recursion formula for the remaining terms is given below. Write out the first ten terms of the sequence.

$$a_1 = 5, a_{n+1} = (-1)^{n+1} a_n / 2$$

Note that the first term of the sequence is given in the problem statement, $a_1 = 5$.

A recursion formula calculates terms of a sequence from terms that precede it. Note that the given recursion formula, $a_{n+1} = (-1)^{n+1} a_n / 2$, uses the previous term of the sequence, a_n , and n to calculate the next term of the sequence, a_{n+1} .

To find the second term of the sequence, let $a_{n+1} = a_2$. Find the value of n .

$$n = 1$$

Thus, substitute $n = 1$ into the recursion formula to solve for a_2 .

$$\begin{aligned} a_2 &= (-1)^{1+1} a_1 / 2 \\ &= (-1)^2 \frac{5}{2} && \text{Substitute } a_1 = 5. \\ &= \frac{5}{2} && \text{Simplify.} \end{aligned}$$

Now find the third term of the sequence. Find the value of n if $a_{n+1} = a_3$.

$$n = 2$$

Thus, substitute $n = 2$ into the recursion formula to solve for a_3 .

$$\begin{aligned} a_3 &= (-1)^{2+1} a_2 / 2 \\ &= (-1)^3 \left(\frac{5}{2} \right) / 2 && \text{Substitute } a_2 = \frac{5}{2}. \\ &= -\frac{5}{4} && \text{Simplify.} \end{aligned}$$

In the same way, substitute $n = 3$ into the recursion formula to solve for a_4 .

$$\begin{aligned} a_4 &= (-1)^{3+1} a_3 / 2 \\ &= (-1)^4 \left(-\frac{5}{4} \right) / 2 && \text{Substitute } a_3 = -\frac{5}{4}. \\ &= -\frac{5}{8} && \text{Simplify.} \end{aligned}$$

Substitute $n = 4$ into the recursion formula to solve for a_5 .

$$\begin{aligned} a_5 &= (-1)^{4+1} a_4 / 2 \\ &= (-1)^5 \left(-\frac{5}{8} \right) / 2 && \text{Substitute } a_4 = -\frac{5}{8}. \\ &= \frac{5}{16} && \text{Simplify.} \end{aligned}$$

Substitute $n = 5$ into the recursion formula to solve for a_6 .

$$\begin{aligned} a_6 &= (-1)^{5+1} a_5 / 2 \\ &= (-1)^6 \left(\frac{5}{16} \right) / 2 && \text{Substitute } a_5 = \frac{5}{16}. \\ &= \frac{5}{32} && \text{Simplify.} \end{aligned}$$

Substitute $n = 6$ into the recursion formula to solve for a_7 .

$$\begin{aligned}
 a_7 &= (-1)^{6+1} a_6 / 2 \\
 &= (-1)^7 \left(\frac{5}{32} \right) / 2 && \text{Substitute } a_6 = \frac{5}{32}. \\
 &= -\frac{5}{64} && \text{Simplify.}
 \end{aligned}$$

Substitute $n = 7$ into the recursion formula to solve for a_8 .

$$\begin{aligned}
 a_8 &= (-1)^{7+1} a_7 / 2 \\
 &= (-1)^8 \left(-\frac{5}{64} \right) / 2 && \text{Substitute } a_7 = -\frac{5}{64}. \\
 &= -\frac{5}{128} && \text{Simplify.}
 \end{aligned}$$

Substitute $n = 8$ into the recursion formula to solve for a_9 .

$$\begin{aligned}
 a_9 &= (-1)^{8+1} a_8 / 2 \\
 &= (-1)^9 \left(-\frac{5}{128} \right) / 2 && \text{Substitute } a_8 = -\frac{5}{128}. \\
 &= \frac{5}{256} && \text{Simplify.}
 \end{aligned}$$

Substitute $n = 9$ into the recursion formula to solve for a_{10} .

$$\begin{aligned}
 a_{10} &= (-1)^{9+1} a_9 / 2 \\
 &= (-1)^{10} \left(\frac{5}{256} \right) / 2 && \text{Substitute } a_9 = \frac{5}{256}. \\
 &= \frac{5}{512} && \text{Simplify.}
 \end{aligned}$$

Thus, the infinite sequence defined by the given recursion formula and value of a_1 is as shown below.

$$5, \frac{5}{2}, -\frac{5}{4}, -\frac{5}{8}, \frac{5}{16}, \frac{5}{32}, -\frac{5}{64}, -\frac{5}{128}, \frac{5}{256}, \frac{5}{512}, \dots$$